

AI & Machine Learning for Civil Engineers

From engineering data to prediction, explainability and uncertainty, in eight hands-on sessions.

08

📅 Live sessions

90 min

🕒 Per session

04 weeks

⌚ Course duration

**Runs 100% on Google Colab**

Nothing to install, works even on a Chromebook

**Built for beginners**

No prior Python or machine learning required

**Live code-along on real data**

Concrete, geotechnical, water and transport datasets

**Notebooks, assignments & solutions**

Plus a certificate and session recordings

INSTRUCTOR**Dr. Abdul Waris**

Research Associate

Universität Siegen (University of Siegen)

Former Prime Minister's Research Fellow

Ph.D., IIT Hyderabad



Research Excellence Award 2026, IIT Hyderabad



Invited talks at Oregon State University (USA), Norway & Spain

Course outline

Problem → Data → Model → Validate → Interpret → Decide

01 Python for civil engineers

AI/ML basics, Colab, NumPy & pandas on concrete data

02 From raw data to ML-ready

Explore, clean & split data; catch outliers and leakage

03 Your first prediction model

Linear regression, error metrics & cross-validation

04 Trees & ensemble models

Decision trees, Random Forest & boosting for nonlinear data

05 Classification for decisions

Logistic regression, confusion matrix, precision & recall

06 Artificial neural networks

Architecture, activations, loss, epochs & overfitting

07 Explainable AI & engineering judgement

Feature importance & SHAP on a geotechnical case study

08 Uncertainty & reliability analysis

Monte Carlo with ML surrogates to estimate failure probability

WHO SHOULD JOIN

B.Tech & M.Tech students / PhD scholars / Faculty
Practising engineers / Complete beginners to AI & ML

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📅 Starts Saturday, 17 October 2026